

UPGRADE — v17.59 EMERGENCY FIX: Inverted Logic During Strong Trends

Severity: EMERGENCY · **DB migration:** ❌ NONE · **Schema change:** ❌ NONE · **Backward compatible:** ✅ YES · **Deploy:** immediately

Summary

v17.59 is an **emergency fix** for inverted entry logic during strong directional trends. The Pine indicator now resolves HTF/LTF trend direction with an explicit **override path**, reports **range-expansion debug** context, and emits an **AMD velocity** reading (as a % of ATR) plus a derived **strong-trend mode**.

The backend update is intentionally **minimal**: it adds **9 OPTIONAL webhook fields** that are **IN-MEMORY ONLY**. They are parsed safely, echoed on `POST /signal`, and surfaced on `GET /latest` — **but they are NOT persisted to the database**. There is **no schema migration and no schema change**, so this release can be deployed immediately with zero database risk.

New Fields (9, all optional, in-memory only)

Field	Type	Domain / Example	Default	Purpose
<code>htf_override_active</code>	bool	0 / 1	false	HTF trend override engaged
<code>ltf_override_active</code>	bool	0 / 1	false	LTF trend override engaged
<code>htf_trend_final</code>	int	-1 / 0 / 1	0	Final resolved HTF trend direction
<code>ltf_trend_final</code>	int	-1 / 0 / 1	0	Final resolved LTF trend direction
<code>range_anchor_time</code>	text	"2026-06-06T09:30:00Z"	""	Debug anchor timestamp of the active range
<code>range_force_expanded</code>	bool	0 / 1	false	The range was force-expanded
<code>amd_velocity</code>	float	87.5 (% of ATR)	0.0	AMD velocity reading
<code>strong_trend_mode</code>	text	BEARISH / BULLISH / NONE	NONE	Derived strong-trend context

Plus: the existing `amd_state` field may be sent with a velocity-based value (e.g. `MARKDOWN`); it continues to be validated/normalized by the existing `normalize_amd_state` path (no behaviour change).

Why in-memory only? These fields are ephemeral real-time context for the frontend HUD (so it can detect “strong trend mode” and avoid inverted entries). They mirror the v17.57 PDH/PDL in-memory pattern — no DB columns, no migration.

Backend Changes

File	Change
<code>src/version.py</code>	<code>VERSION</code> → v17.59 ; added features <code>trend_override_logic</code> , <code>amd_velocity_detection</code> (all existing flags preserved); added <code>normalize_strong_trend_mode</code> helper (domain <code>BEARISH</code> / <code>BULLISH</code> / <code>NONE</code> , default <code>NONE</code>).
<code>src/oie_processor.py</code>	<code>normalize_oie_payload</code> parses the 9 fields (safe parsers) into the record; new <code>apply_trend_override_fields(payload, record)</code> helper; new <code>decode_v17_59_payload</code> ; the helper is also applied (defaults) in every older decoder for backward compat; <code>decode_payload_dispatcher</code> routes <code>version.startswith("v17.59")</code> first.
<code>src/webhook_server/routes.py</code>	<code>_update_latest_signal</code> snapshot includes the 9 fields; <code>POST /signal</code> response echoes the 9 fields; module docstring updated.
<code>app.py</code>	Startup banner mentions “Trend Override + AMD Velocity”.
<code>src/database.py</code>	No change (no migration).

API Behaviour

- **POST /api/v1/signal** — accepts the 9 new fields (optional) and echoes them in the JSON response. Missing/invalid values fall back to safe defaults.
- **GET /api/v1/latest** — the in-memory snapshot now includes the 9 fields.
- **GET /api/v1/health** — reports `version: v17.59` and the 2 new feature flags.
- **GET /api/v1/signals** — **unchanged** (the 9 fields are not persisted, so they do not appear in DB-backed signal rows).

Safe Parsing & Graceful Coercion

All fields use the existing safe parsers (`_to_int` , `_to_float`) and the new `normalize_strong_trend_mode` , so malformed input never crashes the webhook path:

Input	Result
<code>amd_velocity = "not_a_number"</code>	<code>0.0</code>
<code>htf_trend_final = None</code>	<code>0</code>
<code>htf_override_active = "yes"</code>	<code>False</code>
<code>strong_trend_mode = "sideways"</code>	<code>"NONE"</code>
<code>range_force_expanded = ""</code>	<code>False</code>

Backward Compatibility

All prior payload versions (**v17.58 / v17.57 / v17.56.9 / v17.56.8 / v17.56.7 / legacy**) decode exactly as before; the 9 new fields default to `false / 0 / 0.0 / "" / "NONE"`. No existing field, endpoint, or DB row is affected. The signals table schema is unchanged.

Testing

- `test_v17_59.py` — 111 checks, all passing (version flags, 9-field decode, dispatcher routing, backward compat across 6 prior versions, graceful coercion, domain validation, **in-memory-only schema assertion**, `/health`, `POST /signal echo`, `GET /latest surface`).
- Regression: `test_v17_58.py` (155), `test_v17_57.py` (97), `test_v17_56_9.py` (66), `test_v17_56_8.py` (53), `test_oie_integration.py` — all passing.

```
python3 test_v17_59.py
```

Deployment

1. Merge the PR (Railway auto-deploys from `main`).
2. **No migration step** — the DB schema is unchanged.
3. Verify `GET /api/v1/health` reports `v17.59` + the 2 new feature flags.

PR / Merge Order

This release stacks on the existing version chain:

```
#12 → #14 → #16 → #17 → #18 → #19 (v17.59)
```

PR **#19** — `feature/v17.59-emergency-fix` → `feature/v17.58-sequence-state-machine`.
Because this is an emergency fix with no DB risk, it may be fast-tracked to `main` once the upstream chain is merged.